

DevOps Engineering Foundation - 6-Week Intensive Bootcamp

WEEK 2: Networking, Web Servers & Cloud Fundamentals

Learning Outcomes:

- Understand networking basics (TCP/IP, DNS, HTTP/HTTPS)
- Configure and manage web servers (Nginx, Apache)
- Navigate cloud platforms (AWS/Azure fundamentals)
- Deploy basic applications on cloud infrastructure

Resources:

Networking Basics:

- [Computer Networking Course - Network Engineering](#) by freeCodeCamp (9 hours - watch sections on TCP/IP, DNS, HTTP)
- [Networking Basics for DevOps](#) by TechWorld with Nana (30 min)
- [Networking Fundamentals](#) - Cloudflare Learning Center

Web Servers:

- [Nginx Tutorial for Beginners](#) by freeCodeCamp (1 hour)
- [Apache Web Server Tutorial](#) by edureka (30 min)
- [Nginx Beginner's Guide](#)

Cloud Computing (AWS Focus):

- [AWS Certified Cloud Practitioner Certification Course](#) by freeCodeCamp (4 hours - watch fundamentals)
- [AWS EC2 Tutorial for Beginners](#) by Simplilearn (45 min)
- [AWS Free Tier Guide](#)
- [AWS Well-Architected Framework](#) - Fundamentals

Week 2 Project:

Multi-Server Web Application Deployment

Deploy a web application with the following architecture:

1. Launch 2 EC2 instances on AWS
2. Configure Nginx on one instance as a reverse proxy
3. Configure Apache on the second instance as the application server
4. Host a simple static website or Node.js/Python application
5. Configure proper security groups and firewall rules
6. Set up a custom domain (optional: use free services like Freenom) or use Route53
7. Document the entire architecture with diagrams

Deliverables:

- Architecture diagram (use draw.io or Lucidchart)
- Configuration files for both servers
- Public URL of the deployed application
- Detailed deployment documentation
- Security checklist (ports, SSH keys, firewall rules)